

LAB # 4: ER DESIGN CASE STUDY

Objective:

ER Modeling

Scope:

Study and apply ER Modeling in the form of a Case Study.

Tool:

Draw io or Lucid chart or ERD Plus

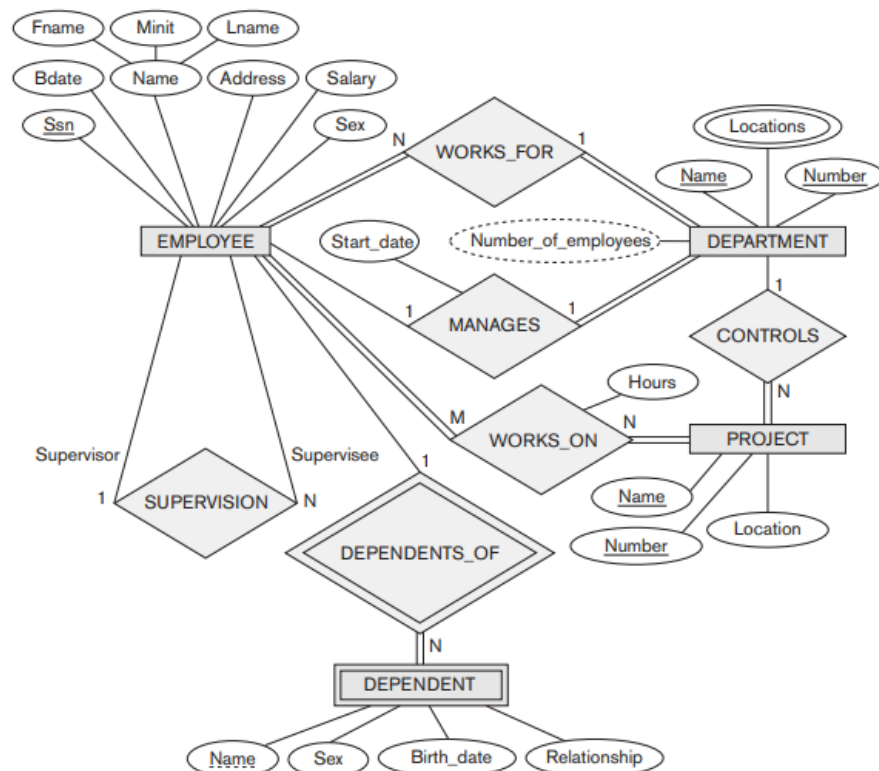
Entity-Relationship Model:

The entity-relationship (ER) model is a popular high-level conceptual data model. This model and its variations are frequently used for the conceptual design of database applications, and many database design tools employ its concepts.

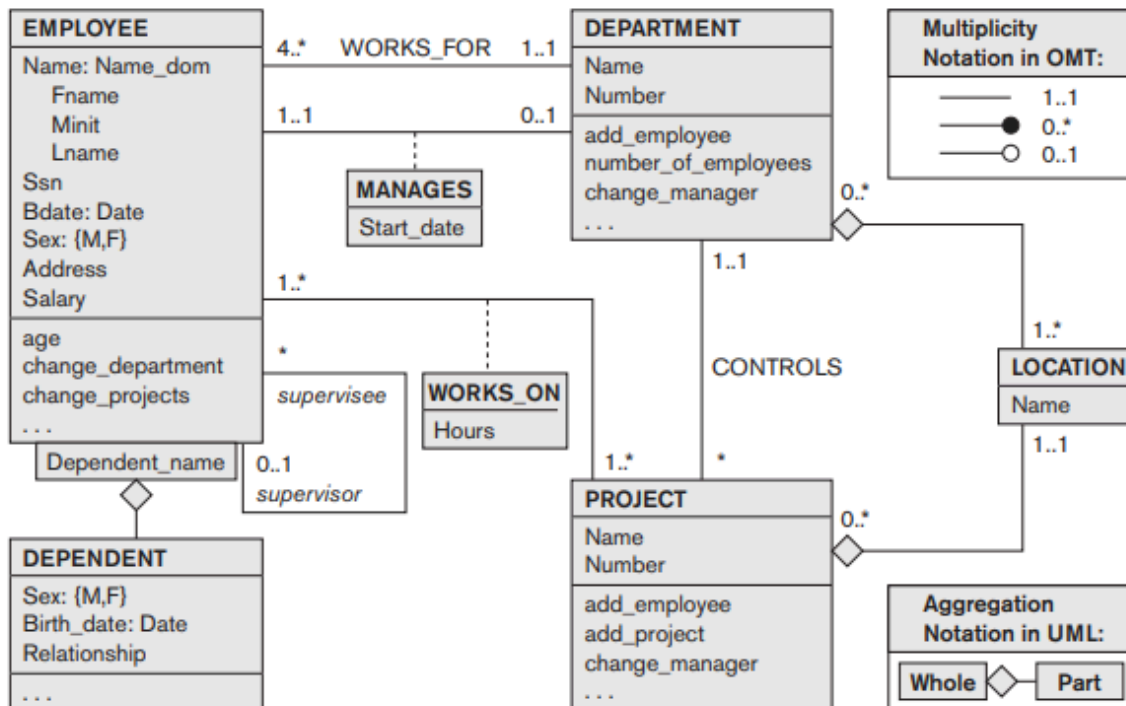
Example:

A sample database application, called COMPANY, serves to illustrate the basic ER model concepts and their use in schema design. We list the data requirements for the database here, and then create its conceptual schema step-by-step as we introduce the modeling concepts of the ER model. The COMPANY database keeps track of a company's employees, departments, and projects. Suppose that after the requirements collection and analysis phase, the database designers provide the following description of the mini world—the part of the company that will be represented in the database.

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- The database will store each employee's name which is a composite attribute, Social Security number, address, salary, sex (gender), and birth date. An employee is assigned to one department but may work on several projects, which are not necessarily controlled by the same department. It is required to keep track of the current number of hours per week that an employee works on each project, as well as the direct supervisor of each employee (who is another employee).
- The database will keep track of the dependents of each employee for insurance purposes, including each dependent's first name, sex, birth date, and relationship to the employee.



Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1: N for $E_1 : E_2$ in R
	Structural Constraint (min, max) on Participation of E in R



How to use the Draw io?

Go to the draw io website by visiting: <https://www.drawio.com/> and click on start



Blog

Start Now

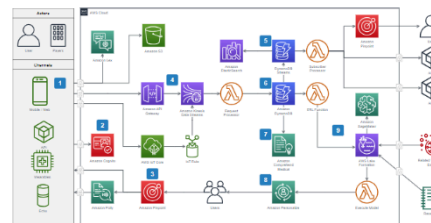
Security-first diagramming for teams.

Bring your storage to our online tool, or save locally with the desktop app.

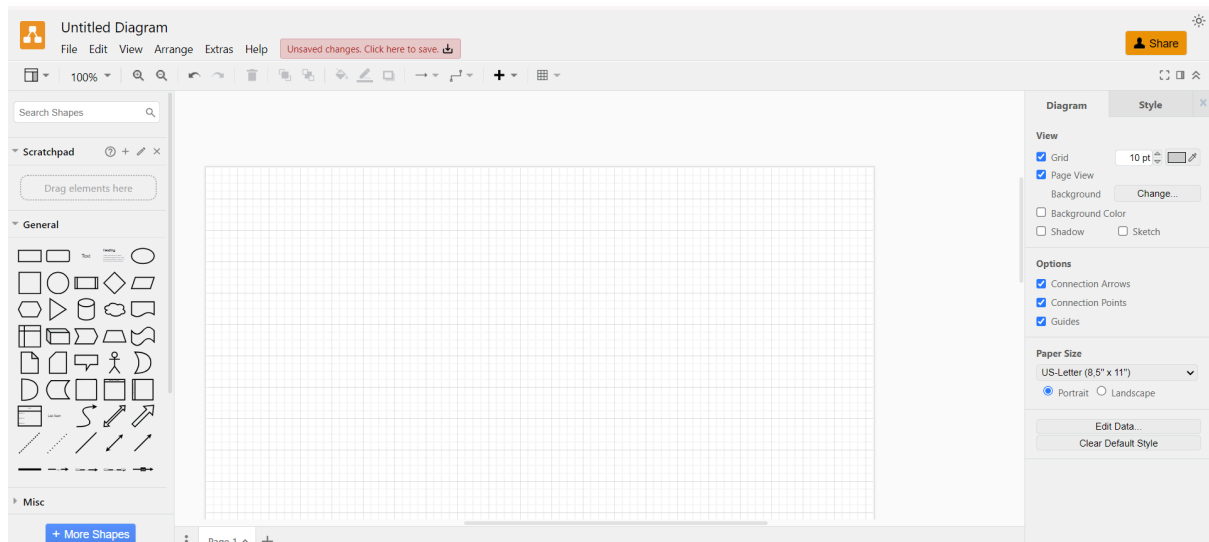
Start

Download

No login or registration required.

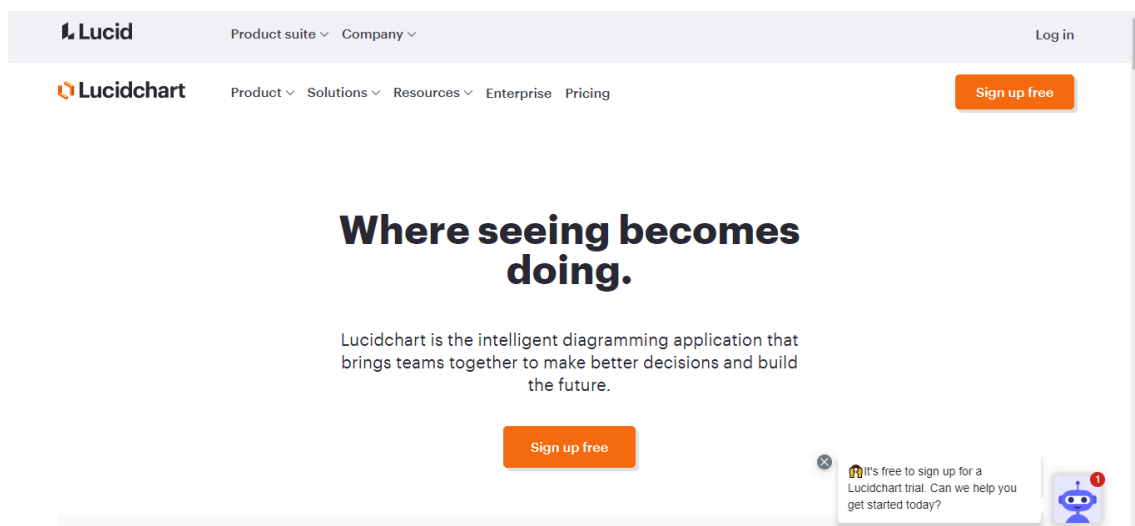


This interface will appear and you can choose Entity Relation option from the left menu to choose relevant shapes for your ER diagram.



How to use the Lucidchart?

Visit the website of Lucidchart by clicking on the link <https://www.lucidchart.com/pages/>, you will see the screen given below.

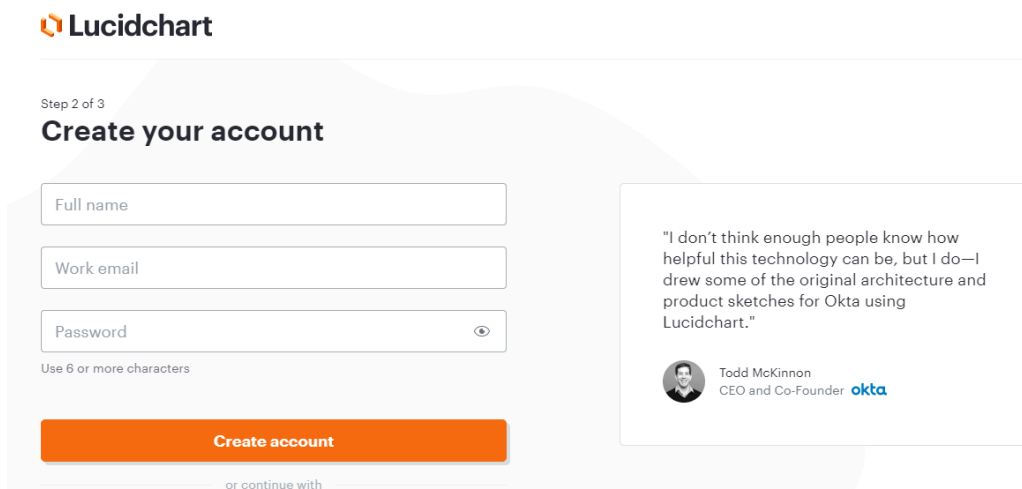


On the top right corner, you can see the button **Sign up Free**, click on that button. After clicking on the button, you will see the pricing plan screen.

We want to use this tool free of cost so we will click on the button at the bottom right corner **Continue with Free**. After clicking on the button, you will see the account creation screen.

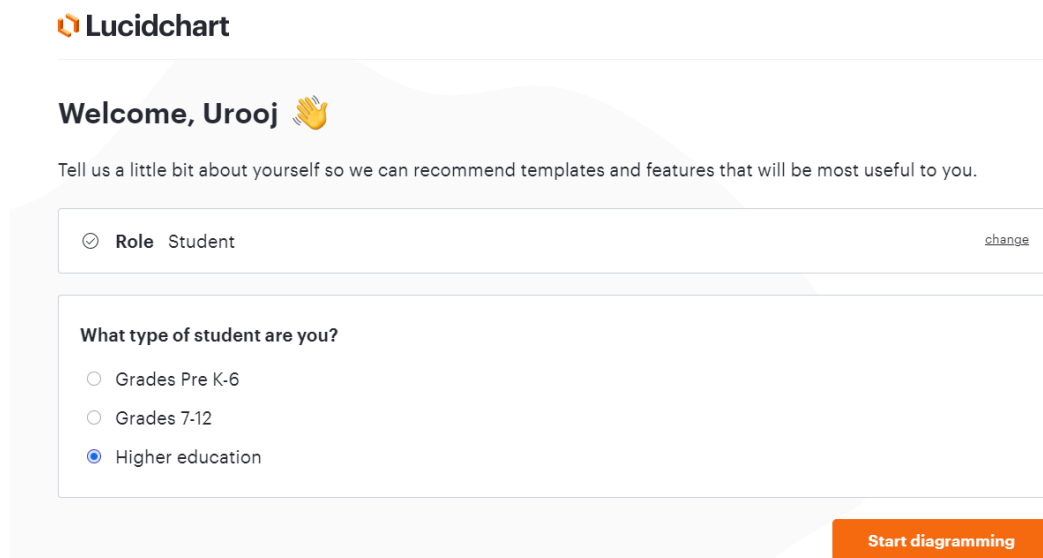
Lucidchart		Contact sales		
	Free	Individual	Team	Enterprise
Price per month	US\$0	Starting at US\$7.95	Starting at US\$9.00 per user 3 user minimum	Contact us for a quote!
Editable documents	3	Unlimited	Unlimited	Unlimited
Professional templates	100	1000+	1000+	1000+
Collaboration	Basic	Basic	Advanced	Advanced
Integrations	Basic	Basic	Advanced	Advanced
Security	✓	✓	✓	✓
Easy admin controls			Basic	Advanced
Advanced data and automation				✓

Enter your full name, university email address, and any password which you want to keep. After entering all the information click on the button **Create account**.



The image shows the 'Create your account' screen on the Lucidchart website. It is labeled 'Step 2 of 3'. The form includes three input fields: 'Full name', 'Work email', and 'Password' (with a toggle for visibility). Below the password field is a note: 'Use 6 or more characters'. An orange 'Create account' button is at the bottom. To the right, there is a quote from Todd McKinnon, CEO and Co-Founder of Okta, and a small profile picture of him.

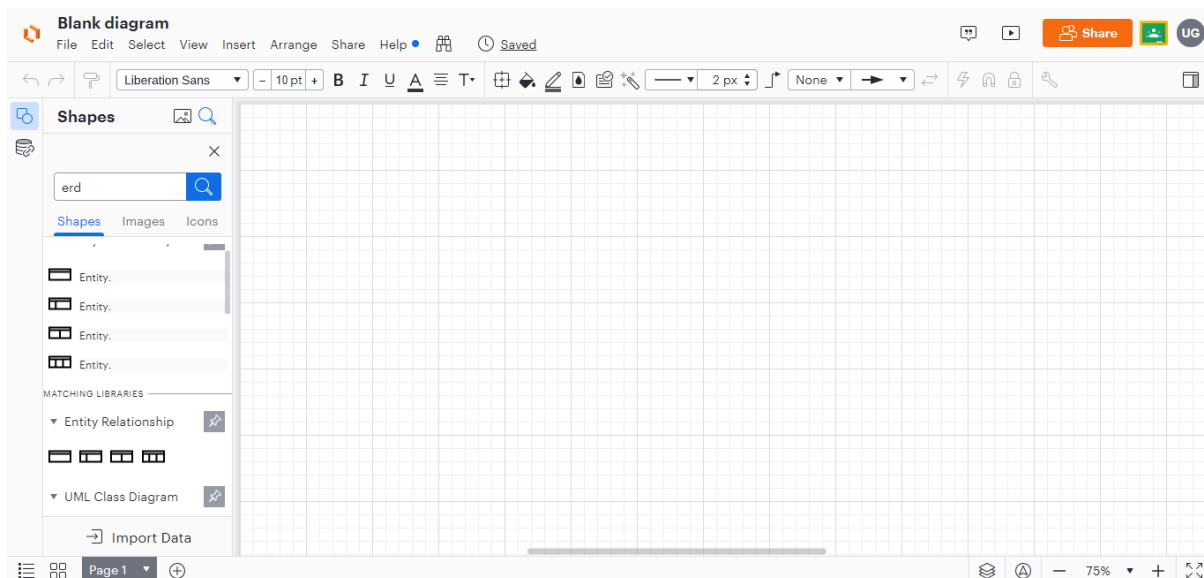
After the successful creation of the account, you will see the screen given below, with some



The image shows the 'Welcome, Urooj' screen on the Lucidchart website. It includes a greeting with a hand icon and a prompt: 'Tell us a little bit about yourself so we can recommend templates and features that will be most useful to you.' Below this is a form with a 'Role' dropdown set to 'Student' and a 'change' link. Underneath is a section titled 'What type of student are you?' with three radio button options: 'Grades Pre K-6', 'Grades 7-12', and 'Higher education' (which is selected). An orange 'Start diagramming' button is at the bottom right.

radio buttons consisting of your role and your education. In role select **Student**, and in education select **Higher Education**. Then click on the button present on the bottom right of the screen **Start diagramming**.

Tada, you have successfully followed all the steps. Now start making diagrams 😊



You can see the blank page; this is the area where you can create your diagram. On the left side of the screen, there is the option **Shapes**, you can write the name of the shape like I have written ERD, so it shows all notations of an ER diagram. Now you can easily drag and drop the notation on the blank page. Once you complete the diagram save it. The option of saving is under the file option.